
DG INTERNS HUB

CYBERSECURITY INTERNSHIP PROGRAMME

WEEK 2

Advanced Networking, Analysis & Security Concepts

Technical Learning & Practical Report

TECHNICAL LEARNING REPORT

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NAVIGATION

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This report documents the theory studied and the practical work performed during Week 2 of the DG Interns Hub Cybersecurity Internship. Every screenshot in the practical sections is original evidence captured during the exercises described.

Week 2 Overview & Objectives

Assignment objective: understand advanced networking concepts, analyze real network traffic, perform basic network scanning, and learn attack concepts conceptually and safely.

Scope: the week combined three theory topics — ports and services, networking devices, and network security basics — with four hands-on practicals: live traffic analysis in Wireshark, network scanning with Nmap, a small LAN build in Cisco Packet Tracer, and a safe, conceptual MITM position simulation.

Tools used

- **Wireshark** — network protocol analyzer, used to capture and inspect live traffic.
- **Nmap** — network scanning tool, used to identify open ports and running services on an authorized target.
- **Cisco Packet Tracer** — network simulator, used to build and test a small LAN and a conceptual MITM topology.

Learning progression

Ports & Services → Networking Devices → Network Security Concepts → Wireshark
→ Nmap → Packet Tracer LAN → Safe MITM Simulation

LEARNING REFLECTION

Week 2 moved my learning from networking theory into practical security analysis. I started by revisiting how ports, devices and basic attack concepts fit together, and then used that theory to make sense of what I was actually seeing on screen — real DNS and TLS traffic in Wireshark, real open ports on an authorized Nmap target, and a small simulated network in Packet Tracer where I could test connectivity myself. By the end of the week, security concepts like MITM and packet sniffing felt less abstract because I had already worked with the traffic and topology they describe.

Ports & Services

A **port** is a transport-layer number used to identify a specific communication endpoint or service on a host. If an IP address gets data to the right *machine*, the port number gets it to the right *application or service* running on that machine.

IP VS PORT

IP address → identifies the host on the network.
Port number → identifies the service or endpoint on that host.

TCP VS UDP

TCP → connection-oriented, reliable, ordered delivery.
UDP → connectionless, lower overhead, no delivery guarantee.

Why ports matter: they allow one IP address to run many services at once. A web server and a mail server on the same machine are separated only by which port each service listens on — which is also why ports are central to both network troubleshooting and security scanning.

Port & Service Reference Table

Port	Protocol	Service	Usage
21	TCP	FTP	File transfer — control/commands channel
22	TCP	SSH	Secure remote login and administration
23	TCP	Telnet	Remote login (unencrypted)
25	TCP	SMTP	Sending / relaying email
53	UDP/TCP	DNS	Domain name resolution
67	UDP	DHCP Server	DHCP server-side communication
68	UDP	DHCP Client	DHCP client-side communication
69	UDP	TFTP	Simple, lightweight file transfer
80	TCP	HTTP	Standard (unencrypted) web communication
110	TCP	POP3	Receiving email
123	UDP	NTP	Time synchronization
143	TCP	IMAP	Email access / synchronization
161	UDP	SNMP	Network device management and monitoring
389	TCP/UDP	LDAP	Directory services
443	TCP	HTTPS	TLS-protected web communication
445	TCP	SMB	Windows file and printer sharing
514	UDP	Syslog	Log / message transport
587	TCP	SMTP Submission	Authenticated email submission
636	TCP	LDAPS	LDAP directory access over TLS
993	TCP	IMAPS	IMAP over TLS
995	TCP	POP3S	POP3 over TLS
3389	TCP/UDP	RDP	Remote Desktop access

SECURITY RELEVANCE

Open ports form part of an exposed attack surface and can be examined by attackers.

KEY INSIGHT

Port identifies the endpoint; TCP/UDP identifies the transport method.

LEARNING REFLECTION

The port table became a practical reference for interpreting service names later in Wireshark and Nmap.

Networking Devices

Three devices came up repeatedly in Week 2: the router, the switch, and the firewall. Each operates at a different point in the network and solves a different problem.

Router

Definition:a device that connects different networks together and forwards IP packets between them. **How it works:** it operates at Layer 3, reading the destination IP address of each packet and deciding which path to send it down. **Real-world example:** a home router connecting a private LAN to the internet. **Key concept:** Layer 3, IP addressing, packets, routing between different networks.

Switch

Definition:a device that connects multiple devices within the same LAN and forwards traffic between them. **How it works:** it operates at Layer 2, using MAC addresses to forward Ethernet frames to the correct local port. **Real-world example:** an office switch connecting PCs and printers on the same floor. **Key concept:** Layer 2, MAC addressing, frames, local connectivity.

Firewall

Definition:a security control that inspects traffic and enforces rules about what is allowed to pass. **How it works:** it inspects traffic against a defined rule set and allows or denies it accordingly. **Real-world example:** a company firewall controlling traffic between the internet and the internal network. **Key concept:** traffic inspection, security rules, allow/deny decisions.

Router vs Switch

	Router	Switch
Scope	Connects different networks	Connects devices on the same/local network
Addressing	IP address	MAC address
Unit forwarded	Packets	Frames
Function	Routing between networks	Local connectivity

KEY MEMORY

Switch connects. Router routes. Firewall controls.

LEARNING REFLECTION

Separating the three device roles made later practicals easier to reason about.
The switch explained same-LAN connectivity, the router became relevant beyond the local network, and the firewall added the security-control perspective.

04 / THEORY — TASK 3

Network Security Basics

Network security is the practice of protecting the confidentiality, integrity and availability of data as it moves across a network. Task 3 focused on understanding three foundational concepts conceptually and safely, without performing any attack.

Man-in-the-Middle (MITM)

Definition: a MITM attack occurs when an attacker positions themselves between two communicating parties and may observe or interfere with their communication.

Victim → Attacker → Server

Real-world example: public Wi-Fi, where an attacker may be able to position themselves between a victim's device and the intended destination. **Risks:** data theft, password capture, communication interception. **Basic prevention:** HTTPS, VPN, and using only secure/trusted networks.

Denial of Service (DoS)

Definition: a DoS attack attempts to make a system, service, or network resource unavailable to legitimate users.

Excessive / Malicious Traffic → Resource Exhaustion → Service Disruption

Main impact: availability. **Risks:** service unavailability, performance degradation, legitimate users unable to access the service. **Basic prevention:** traffic filtering, rate limiting, firewalls, and network monitoring.

Packet Sniffing

Definition: the process of capturing and examining network packets as they travel across a network.

Network Traffic → Packets → Captured → Examined

What can be observed: source/destination IP, protocol, ports, and — depending on encryption — application-level content. **Basic prevention:** HTTPS/encryption, VPN, secure/trusted networks. This is exactly the concept exercised safely in Task 4 using Wireshark.

Comparison

Concept	Main Idea	Main Security Impact	Basic Prevention
MITM	Attacker positioned in the communication path	Confidentiality / integrity	HTTPS, VPN, trusted networks
DoS	Resources overwhelmed by traffic	Availability	Filtering, rate limiting, firewalls
Packet Sniffing	Traffic captured and examined	Confidentiality	Encryption (HTTPS/VPN)

KEY POINT

Encrypted traffic can still be captured, but encryption protects the contents from being easily read by an observer — a distinction I confirmed directly in Task 4 when comparing a plain DNS query to an encrypted TLS session.

LEARNING REFLECTION

MITM is about position, packet sniffing is about observation, and DoS is about availability rather than confidentiality. Keeping these three ideas conceptually separate made it much easier to understand why the same basic prevention — HTTPS and trusted networks — shows up as a defense against more than one of them.

05 / PRACTICAL — TASK 4

Wireshark Packet Analysis

Wireshark is a network protocol analyzer used to capture and inspect network packets in real time. It helps identify the source IP (the sender), destination IP (the receiver), protocol (the type of communication), and port (the specific service/endpoint involved) for every packet on the wire.

Reading a single packet means working through several layers of detail: the **packet list** (a one-line summary of every frame captured), the **packet details** pane (the IP layer, the TCP/UDP layer, and any application-specific information such as DNS or TLS fields), and the raw bytes underneath. For this practical, the focus was on the IP addresses, the transport-layer port, and the protocol identified for each packet.

Step 1 — Select the Correct Interface | Step 2 — Generate Traffic

The active **Wi-Fi 2** interface was selected in Wireshark before starting the capture. While the capture was running, the browser was opened and <https://google.com> was visited to generate normal, real web traffic to analyze.

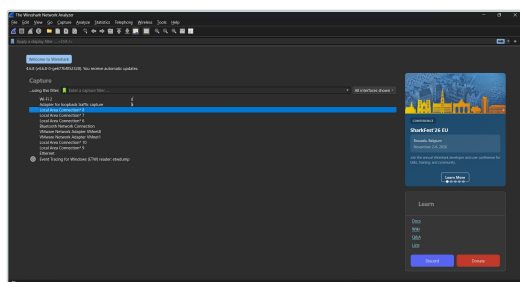


Figure 1 — Active Wi-Fi 2 adapter selected before starting capture.

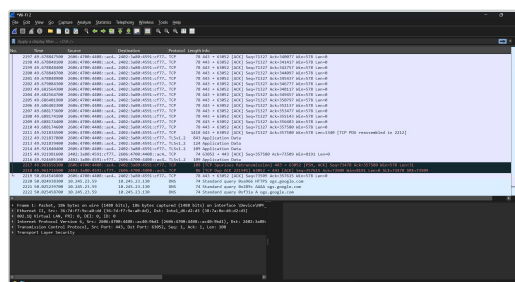


Figure 2 — Live capture on Wi-Fi 2 while browsing generated TCP, TLS and DNS traffic.

PRACTICAL OBSERVATION: The unfiltered capture showed a mix of TCP acknowledgment traffic, TLSv1.2 application data, and DNS queries/responses — confirming that real browsing activity was being recorded on the selected interface.

Step 3 — DNS Analysis | Step 4 — HTTPS/TLS Analysis

The **dns** and **tls** display filters were applied in turn to isolate name-resolution traffic and encrypted web traffic from the same browsing session.

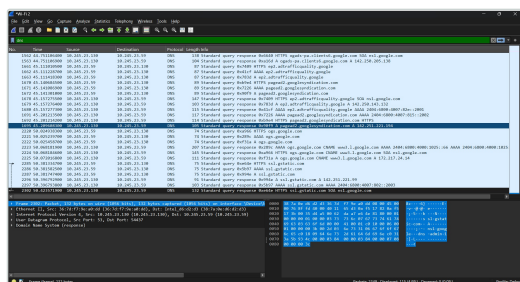


Figure 3 — DNS-filtered capture: queries/responses for domains contacted while loading google.com.

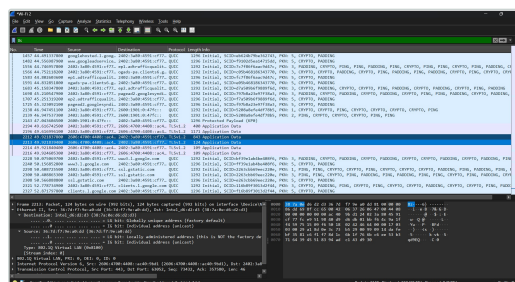


Figure 4 — TLS-filtered capture: a TLSv1.2 Application Data packet from the encrypted session.

PRACTICAL OBSERVATION: A DNS response was identified using **UDP port 53** (source IP **10.245.23.130**, destination IP **10.245.23.59**). Separately, a **TLSv1.2 Application Data** packet was identified on **source port 443**, confirming TLS-protected web communication — the IP addresses were visible at the packet-header level, but the encrypted payload itself was not readable in plain text.

KEY INSIGHT

Encrypted traffic can still be captured, but encryption protects the contents from being easily read. The DNS query in Step 3 was fully readable, while the TLS session in Step 4 exposed only connection metadata — a direct, hands-on confirmation of the theory covered in Task 3.

LEARNING REFLECTION

Working through DNS and TLS traffic side by side showed me the practical difference between plaintext and encrypted protocols. I also learned to navigate Wireshark's filter bar to isolate exactly the traffic I needed instead of scrolling through an unfiltered capture.

06 / PRACTICAL — TASK 5

Nmap Network Scanning

Nmap (Network Mapper) is a network-scanning tool used to identify reachable ports, their states, and the services running behind them on an authorized target. Every scan works through the same basic model:

Port → State → Service

PORT STATES

Open — a service is accepting connections.

Closed — no service is listening.

Filtered — state cannot be determined; traffic is being filtered or blocked.

SERVICE/VERSION DETECTION

The `-sV` option goes a step further than a basic scan — it attempts to identify the actual service and version information running behind each open port.

The official commands for this task were:

```
nmap scanme.nmap.org # basic port and service scan
nmap -sV scanme.nmap.org # service and version detection
```

PRACTICAL NOTE — TIMING ADAPTATION

The exact official command `nmap scanme.nmap.org` was attempted first, but in this environment it took an unusually long time due to repeated retransmissions and did not complete within a reasonable period. To complete the practical on the same authorized target, `-T4` timing optimization was added: `nmap -T4 scanme.nmap.org` and `nmap -sV -T4 scanme.nmap.org`. This is disclosed transparently below rather than presented as the unmodified assignment command.

Practical 1 — Basic Scan

```
nmap -T4 scanme.nmap.org
```

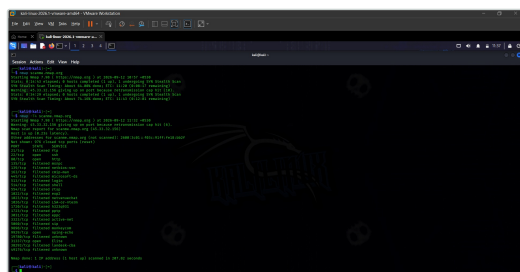


Figure 5 — The original `nmap scanme.nmap.org` stalling on retransmissions, followed by the completed `-T4` scan.

Practical 2 — Service/Version

```
nmap -sV -T4 scanme.nmap.org
```

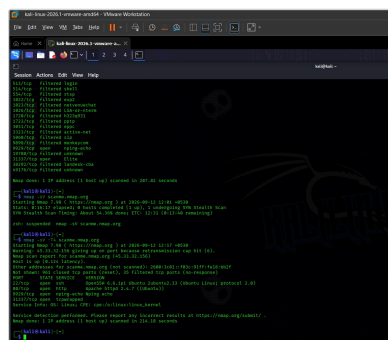


Figure 6 — Service/version detection output showing the software behind each open port.

WHAT I OBSERVED: The scan identified `scanme.nmap.org` as an active Linux host. Ports **22/tcp (SSH)**, **80/tcp (HTTP)**, **9929/tcp (nping-echo)** and **31337/tcp (Elite)** were found **open**, while a number of other commonly-probed ports were reported as **filtered**. Version detection then identified **OpenSSH 6.6.1p1 (Ubuntu Linux)** on port 22 and **Apache httpd 2.4.7 (Ubuntu)** on port 80.

WHY -T4 WAS USED, AND WHAT -SV ADDED

`-T4` is a timing template that increases scan speed without changing which ports are tested — it was used only to make the scan complete reliably, not to change the scope of the assignment. `-sV` went beyond a basic open/closed/filtered result and identified the actual software and version running on the open ports, which is far more useful from a security standpoint than the port list alone.

LEARNING REFLECTION

A scan result depends on real network conditions, not just the command typed — retransmissions and timing can matter as much as the target itself. I also learned that version detection turns a plain port list into something a security analyst can actually act on, since a specific outdated service version is far more actionable than "port 80 is open."

Packet Tracer LAN Setup

A **LAN** connects devices within a local network. A switch provides Layer 2 connectivity between those devices, forwarding Ethernet frames using MAC addresses rather than routing between networks. Because both PCs in this exercise sit in the same /24 subnet, they can communicate directly through the switch — no router is required for this kind of same-network communication.

IP Configuration

Device	IP Address	Subnet Mask
PC1	192.168.1.10	255.255.255.0
PC2	192.168.1.11	255.255.255.0

Evidence

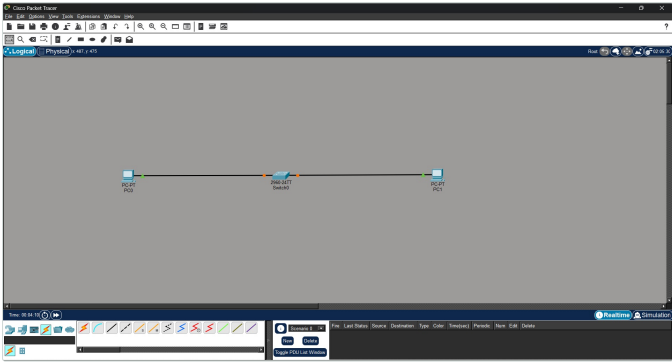
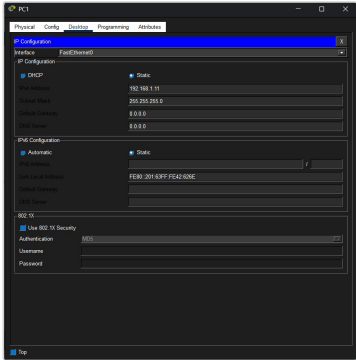
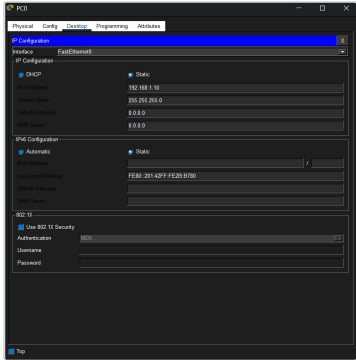


Figure 7 — LAN topology: PC1 and PC2 connected through a single 2960-24TT switch.



Connectivity Verification

```
ping 192.168.1.10
```

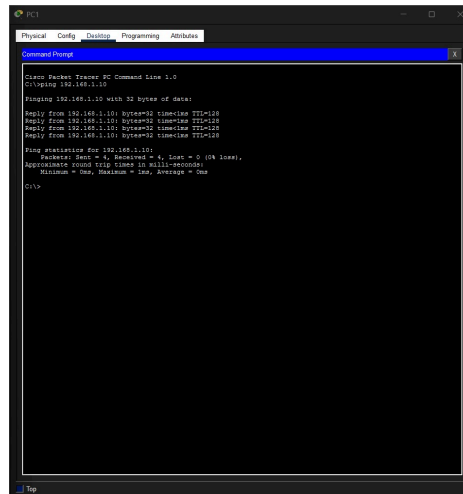


Figure 10 - PC2 pinging PC1 (192.168.1.10) through the switch: 4 packets sent, 4 received, 0% loss.

PRACTICAL OBSERVATION

PC2 successfully pinged PC1 at 192.168.1.10 through the switch. All 4 packets were received with 0% loss, confirming working Layer 2 communication between the two PCs on the same LAN.

LEARNING REFLECTION

Both PCs were placed in the same /24 subnet, so MAC-based Layer 2 forwarding through the switch was sufficient for local communication. This practical made the switch theory directly visible in a working LAN.

Safe MITM Concept Simulation

A **Man-in-the-Middle (MITM)** situation occurs when an attacker positions themselves between two communicating parties and may potentially observe or interfere with their communication. This task used a safe Packet Tracer topology to understand *where* an attacker sits in the communication path — without performing any real interception.

NORMAL COMMUNICATION

PC1 → Switch → Router

CONCEPTUAL MITM POSITION

PC1 → Attacker → Router

Real-world example: a public Wi-Fi network can create a risk of MITM attacks if an attacker is able to position themselves between a user's device and the intended destination. **Risks:** data theft and password capture. **Prevention:** HTTPS, VPN, and secure/trusted networks.

Practical Topology

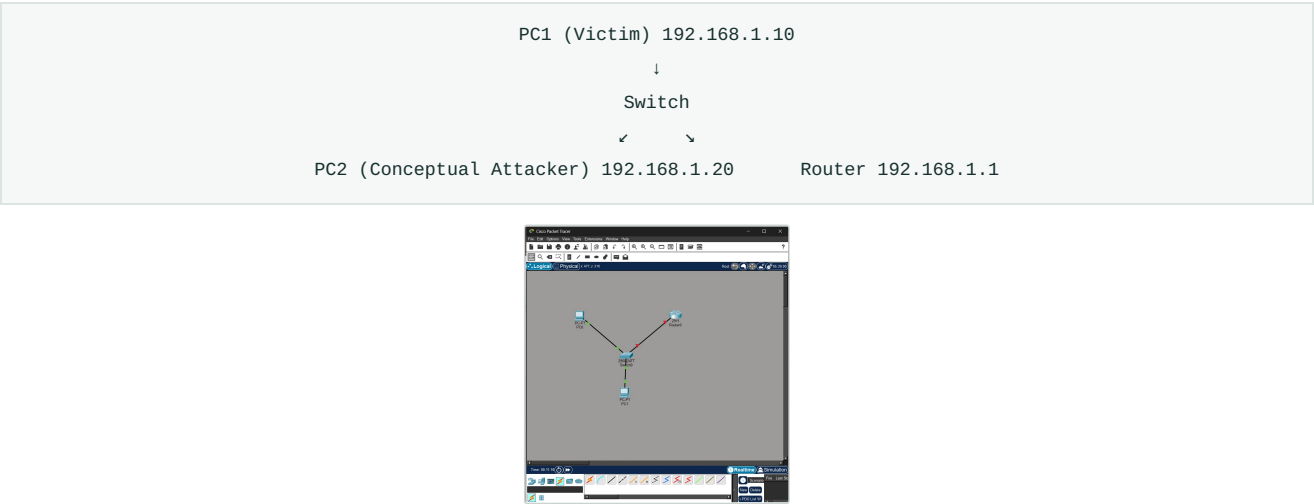
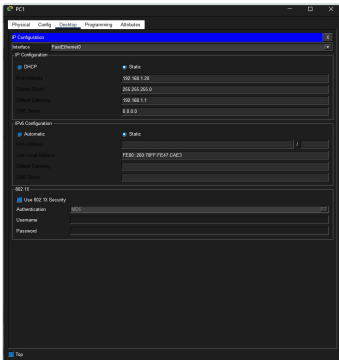
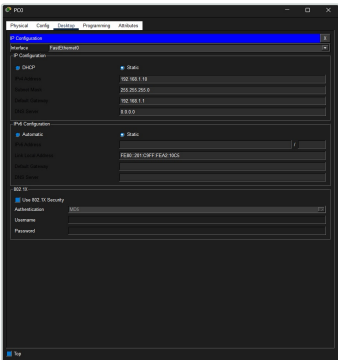


Figure 11 — Packet Tracer topology: PC1 (victim) and PC2 (conceptual attacker) both connected through a switch to the router.

Step 1 — IP Configuration

Device	IP Address	Subnet Mask	Default Gateway
PC1 (Victim)	192.168.1.10	255.255.255.0	192.168.1.1
PC2 (Attacker)	192.168.1.20	255.255.255.0	192.168.1.1
Router	192.168.1.1	255.255.255.0	—



Step 2 - Test Normal Connectivity

```
ping 192.168.1.1
```

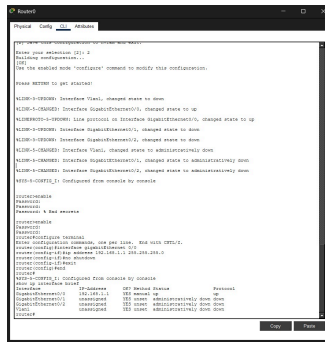


Figure 14 - Router CLI: GigabitEthernet0/0 set to 192.168.1.1/24 and verified up/up.

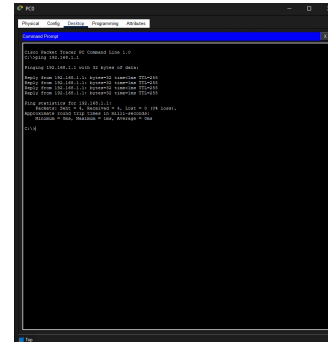


Figure 15 - PC1 pinging the router (192.168.1.1): 4 sent, 4 received, 0% loss.

PRACTICAL OBSERVATION

PC1 successfully pinged the router at 192.168.1.1, receiving 4 out of 4 replies with 0% packet loss. This confirmed normal connectivity between the victim PC, the switch, and the router.

SAFETY BOUNDARY

No ARP poisoning, credential capture, real interception, password theft, or traffic manipulation was performed. This was a safe conceptual Packet Tracer simulation of the attacker position in a communication path.

LEARNING REFLECTION

The topology made the MITM concept concrete: I could see the victim, the conceptual attacker, and the router in relation to one another, while the baseline ping confirmed that normal communication worked first.

Integrated Week 2 Learning

Integrated Week 2 Learning

From networking fundamentals to practical security analysis

01	02	03	04	05	06	07
Ports & Services	Networking Devices	Security Concepts	Wireshark	Nmap	Packet Tracer	Safe MITM

What the week connected

Week 2 connected concepts that I had previously studied separately. Ports and services gave me a vocabulary for network communication, the router, switch and firewall explained how traffic is moved or controlled, and the security topics gave those networking ideas a clear defensive purpose. The practical tasks then turned those ideas into evidence that I could observe, interpret and verify myself.

OBSERVATION

Wireshark changed my learning from theory to evidence. I could identify DNS and TLS traffic directly, read packet headers, and use filters to isolate the traffic that mattered.

ASSESSMENT

Nmap showed how service exposure can be measured in practice. An open port became more meaningful once I connected it to its service and, with -sV, to the software/version behind it.

SIMULATION

Packet Tracer let me build the network myself. I could verify same-subnet communication, router connectivity, and the safe conceptual position of a MITM rather than only describing them in theory.

LEARNING REFLECTION

By the end of Week 2, I was no longer treating networking tools as separate utilities. I could move from a port number to its service, from a packet to the protocol carrying it, from a scan result to the service behind an open port, and from a topology to the path that traffic would normally follow.

The most useful lesson was learning to interpret evidence rather than simply collect it. Wireshark showed what was happening on the network, Nmap showed what was exposed on an authorized host, and Packet Tracer showed how addressing and device roles produced real connectivity. Together, those practicals made the security concepts in Week 2 more concrete and gave me a stronger foundation for future SOC and Blue Team learning.

10 / REFLECTION

Final Reflection & Conclusion

LEARNING REFLECTION

- How ports, TCP/UDP, and common services fit together, and why that vocabulary matters for security tools.
- The distinct roles of a router, switch, and firewall, summarized by "switch connects, router routes, firewall controls."
- The conceptual mechanics of MITM, DoS, and packet sniffing, along with their basic preventions.
- How to capture and filter live traffic in Wireshark, and how to read DNS and TLS packets specifically.
- How to run and interpret an Nmap scan, including the difference a basic scan and a service/version scan make.
- How to build and verify a small LAN, and a slightly larger topology, in Cisco Packet Tracer.

Practical Skills Gained

- Selecting the correct capture interface and applying display filters in Wireshark.
- Running Nmap scans against an authorized target and adjusting timing options when needed.
- Configuring static IPv4 addressing on end devices and router interfaces in Packet Tracer.
- Verifying connectivity with `ping` and reading the resulting statistics.

Challenges Encountered

The default Nmap scan (`nmap scanme.nmap.org`) took unusually long to complete in my environment, stalling on repeated retransmissions well past a reasonable wait time.

How I Adapted

Rather than assuming the target was unreachable, I investigated the behavior, confirmed the target was responding, and used `-T4` timing optimization to complete both the basic scan and the service/version scan on the same authorized target — documenting the change honestly instead of presenting the adjusted command as the original one.

Final Takeaway

Week 2 connected networking theory to visible, practical evidence. Ports and devices explained what I was looking at, Wireshark and Nmap let me observe real traffic and real exposed services, and Packet Tracer let me build and verify a small network myself — including a safe simulation of where a security threat like MITM actually sits. The combination made abstract security concepts noticeably more concrete than theory alone would have.

This report reflects the work actually completed for Week 2 of the DG Interns Hub Cybersecurity Internship. All screenshots are original evidence captured during the practical exercises described above.